HR Analytics Dashboard | Project documentation

1. Problem statement

HR stakeholders need a consistent way to examine workforce composition, employee attrition, compensation, job satisfaction and tenure across departments and roles. A row-level extract makes it difficult to compare groups, identify patterns and assess whether workforce KPIs use consistent denominators. This project uses Power BI to turn an employee-level CSV into an interactive analytical dashboard for descriptive workforce monitoring. It is not a causal attrition model.

2. Project objectives

- Measure employee records, distinct employees and attrition flags.
- Compare attrition proportions across departments, job roles, age groups, salary bands and overtime categories.
- Explore job satisfaction, performance, income and tenure measures.
- Allow viewers to filter report visuals using slicers and cross-highlighting.
- Document source-quality limitations before interpreting or publishing results.

3. Dataset and report inventory

The CSV has 1,480 rows and 38 fields. There are 1,470 unique EmpID values, 10 repeated employee-ID occurrences, 7 exact duplicate rows and 57 missing YearsWithCurrManager values. The Power BI report definition contains 15 visual objects: areaChart (1), card (6), clusteredBarChart (2), columnChart (1), donutChart (1), pivotTable (1), slicer (1), textbox (1), treemap (1).

4. Headline descriptive measures

Measure	Value
CSV records	1,480
Distinct EmpID	1,470
Attrition = Yes	238
Attrition share of rows	16.08%
Average age	36.92 years
Average monthly income	6,504.99 (currency unknown)

5. Data terminology

Field	Operational meaning / caution
EmpID	Employee identifier; duplicates exist, so verify row grain before distinct headcount.
Age	Employee age in years.
AgeGroup	Categorized age band.
Attrition	Yes/No departure indicator; snapshot status, not dated turnover events.

BusinessTravel	Frequency of work-related travel.
DailyRate	Source-defined daily rate; do not assume salary basis.
Department	Organizational department.
DistanceFromHome	Numeric commuting-distance measure; unit not stated.
Education	Coded education level; mapping requires source confirmation.
EducationField	Field of education.
EmployeeCount	Record-level count field, uniformly 1 in this extract.
EmployeeNumber	Numeric employee identifier.
EnvironmentSatisfaction	Coded workplace-environment satisfaction score.
Gender	Gender category in source.
HourlyRate	Source-defined hourly rate; currency not documented.
JobInvolvement	Coded job-involvement score.
JobLevel	Coded job level.
JobRole	Employee job role.
JobSatisfaction	Coded job-satisfaction score.
MaritalStatus	Recorded marital status.
MonthlyIncome	Monthly income; currency not documented.
SalarySlab	Income range category.
MonthlyRate	Source-defined monthly rate; not necessarily MonthlyIncome.
NumCompaniesWorked	Number of previous companies worked.
Over18	Age threshold flag (uniform Y).
OverTime	Yes/No overtime flag.
PercentSalaryHike	Reported percentage salary increase.
PerformanceRating	Coded performance rating.
RelationshipSatisfaction	Coded relationship-satisfaction score.
StandardHours	Source standard-hours field (constant 80).
StockOptionLevel	Coded stock-option level.
TotalWorkingYears	Total career tenure in years.
TrainingTimesLastYear	Number of training sessions in the preceding year.
WorkLifeBalance	Coded work-life balance score.
YearsAtCompany	Years employed at current company.
YearsInCurrentRole	Years in current role.
YearsSinceLastPromotion	Years since last promotion.
YearsWithCurrManager	Years with current manager; 57 missing values.

6. KPI definitions

- Employee records = COUNTROWS(HR_Analytics); use only when each row represents one person.
- Distinct employees = DISTINCTCOUNT(HR_Analytics[EmpID]); duplicates require investigation.
- Attrition count = count of rows where Attrition = "Yes".
- Attrition share = attrition-Yes rows / all rows in the selected filter context. This is not a time-based turnover rate.
- Average age = arithmetic mean of Age across selected rows.
- Average monthly income = arithmetic mean of MonthlyIncome; currency unknown.
- Average tenure = arithmetic mean of YearsAtCompany, excluding blank values.

7. Quality, privacy and limitations

- Duplicate IDs and seven fully duplicated rows can distort headcount and subgroup comparisons.
- The CSV has no hire, termination or snapshot date. Avoid annual turnover claims or month-over-month trends.
- 57 YearsWithCurrManager values are missing. Validate whether blanks have a defined meaning.
- Salary currency, distance units and satisfaction scale labels are not documented.
- EmpID and EmployeeNumber are identifiers; publish a synthetic or properly de-identified dataset.
- Power BI measures inside the binary model were not verified as exact source DAX; tutorial formulas are documented reproductions.

8. Tools and scope

Microsoft Power BI Desktop, Power Query, DAX, report visuals and slicers. Microsoft Excel can open the CSV for inspection, but the supplied project is a PBIX report, not an Excel VBA dashboard.